LOG OF BORING NO.: MB-1

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 1 OF 1
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 1/30/2024 END: 1/30/2024
NORTH: 5239136.236 | EAST: 726696.939 | COORD. DATUM: UTM21-North | LOGGER: Dylan Brown
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ ___ ℄ CAVED @ ___ TIME: ___ DATE: ___ DESC.: ___ PLUNGE: -90
▼ H_2O @ ___ ℄ CAVED @ ___ TIME: ___ DATE: ___ DESC.: ___ ELEV. DATUM: CGVD2013
▼ H_2O @ ___ ℄ CAVED @ ___ TIME: ___ DATE: ___ DESC.: ___ GROUND ELEV.: 14.9

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
0.0	S-1	0%	19 21 25			0	Not described due to no recovery. New sample retainer baskets applied after first sample with no improvement.	
1.5	S-2	0%	24 15 15			1		
3.0	S-3	0%	4 4 4			2		
4.5	S-4	0%	8 4 3			3		
6.0						4	6.0' - EL 8.9	
						5	End of Boring at 6.0'	
						6		
						7		
						8		
						9		
						10		
						11		
						12		
						13		
						14		
						15		
						16		
						17		
						18		
						19		
						20		

Boring terminated due to low recovery in attempt to recover more soil for environmental testing. Boring backfilled with cement-bentonite grout.

TEMPLATE=10L_BORING PROJECT=MULPOINT SMIG PRI LIBRARY GPJ LIBRARY=BAKER 10-1 GLB DATE PRINTED= 3/18/24 S.O.#=197889 CHECKED BY: JML

LOG OF BORING NO.: MB-1A

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 1 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 1/30/2024 END: 2/2/2024
NORTH: 5239136.236 | EAST: 726697.1227 | COORD. DATUM: UTM21-North | LOGGER: Dylan Brown
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 11 ℄ CAVED @ ___ TIME: 12:10 DATE: 2/2/2024 DESC.: 0 Hr. PLUNGE: -90
▼ H_2O @ 11.4 ℄ CAVED @ ___ TIME: 13:30 DATE: 2/3/2024 DESC.: 25.5 Hr. ELEV. DATUM: CGVD2013
▼ H_2O @ ___ ℄ CAVED @ ___ TIME: ___ DATE: ___ DESC.: ___ GROUND ELEV.: 14.9

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
0.0	S-1	67%	5 7 14			0	TOPSOIL. 0.2' - EL 14.7	FILL.
1.5	S-2	13%	8 20 6			1	Clayey SAND with gravel (sc); sand is fine to coarse; gravel is fine to coarse; angular; nonplastic; gray; moist; NP; loose to very dense; heterogeneous. Evidence of cobbles. Petroleum odor in S-6.	
3.0	S-3	0%	9 5 5			2		
4.5	S-4	27%	7 5 3	S-4(4.5-6.0) PID=0 ppm		3		
6.0	S-5	0%	10 9 5			4		
7.5	S-6	7%	6 5 7	S-6(7.5-9.0) PID=30.3 ppm		5	7.5' - EL 7.4	
9.0	S-7	100%	50/0.4	S-7(9.0-9.4) PID=0 ppm		6	Well-graded GRAVEL with silt and sand (gw-gm); sand is fine to coarse; gravel is fine to coarse; subrounded; nonplastic; gray, brownish; moist to wet; NP; very dense; heterogeneous	FILL.
9.4	A-N					7		
10.5	S-8	27%	15 16 13	S-8(10.5-12.0) PID=0 ppm		8	Concrete and rebar recovered around 10.0'	
12.0	S-9	67%	34 21 21	S-9(12.0-13.5) PID=183 ppm; S-9 to S-13(12.0-19.5) PL=NP MC=6.8 USCS=GP-GM P#200=7.9		9	12.0' - EL 2.9	
13.5	S-10	25%	15 26 50/0.2			10	Poorly graded GRAVEL with silt and sand (GP-GM); sand is fine to coarse; gravel is fine to coarse; subrounded; nonplastic; gray, brownish; wet; NP; medium dense to very dense; heterogeneous	GLACIAL.
14.7	A-N					11		
15.0	S-11	50%	29 50/.1			12		
15.6	A-N					13		
16.5	S-12	27%	28 17 14			14		
18.0	S-13	20%	13 15 16			15		
19.5						16		
						17		
						18		
						19		
						20		

Boring backfilled with cement-bentonite grout.

TEMPLATE=10L_BORING PROJECT=MULPOINT SMIG PRI LIBRARY GPJ LIBRARY=BAKER 10-1 GLB DATE PRINTED= 3/18/24 S.O.#=197889 CHECKED BY: JML

LOG OF BORING NO.: MB-1A

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 2 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 1/30/2024 END: 2/2/2024
NORTH: 5239136.236 | EAST: 726697.1227 | COORD. DATUM: UTM21-North | LOGGER: Dylan Brown
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 11 ℄ CAVED @ ___ TIME: 12:10 DATE: 2/2/2024 DESC.: 0 Hr. PLUNGE: -90
▼ H_2O @ 11.4 ℄ CAVED @ ___ TIME: 13:30 DATE: 2/3/2024 DESC.: 25.5 Hr. ELEV. DATUM: CGVD2013
▼ H_2O @ ___ ℄ CAVED @ ___ TIME: ___ DATE: ___ DESC.: ___ GROUND ELEV.: 14.9

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
	S-14	0%	10 17 17			20		
21.0	A-N			Bag 1(21.0-61.0) PL=NP MC=17.8 USCS=SP-SM P#200=9.7		21		
						22		
						23		
						24		
						25		
26.0	S-15	47%	14 23 22	S-15/S-16(26.0-32.5) PL=NP MC=10.8 USCS=SP-SM P#200=12.6		26	26.0' - EL -11.1 Poorly graded SAND with silt and gravel (SP-SM); sand is fine to coarse; gravel is fine to coarse; subangular; nonplastic; gray, greenish; wet; NP; medium dense to dense; homogeneous.	GLACIAL.
27.5	A-N					27		
						28		
						29		
31.0	S-16	53%	14 28 16			30		
32.5	A-N					31		
						32		
						33		
						34		
						35		
36.0	S-17	0%	4 10 13			36		
37.5	A-N					37		
						38		
						39		
						40		

Boring backfilled with cement-bentonite grout.

TEMPLATE=10L_BORING PROJECT=MULPOINT SMIG PRI LIBRARY GPJ LIBRARY=BAKER 10-1 GLB DATE PRINTED= 3/18/24 S.O.#=197889 CHECKED BY: JML

NOTES

1. 1 METER = 3.28 FEET

LOG OF BORING NO.: MB-1A

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 3 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 1/30/2024 END: 2/2/2024
NORTH: 5239136.236 | EAST: 726697.1227 | COORD. DATUM: UTM21-North | LOGGER: Dylan Brown
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 11 ℄ CAVED @ ___ TIME: 12:10 DATE: 2/2/2024 DESC.: 0 Hr. PLUNGE: -90
▼ H_2O @ 11.4 ℄ CAVED @ ___ TIME: 13:30 DATE: 2/3/2024 DESC.: 25.5 Hr. ELEV. DATUM: CGVD2013
▼ H_2O @ ___ ℄ CAVED @ ___ TIME: ___ DATE: ___ DESC.: ___ GROUND ELEV.: 14.9

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
41.0	S-18	27%	8 15 21			40		
42.5	A-N					41		
						42		
						43		
						44		
						45		
46.0	S-19	67%	12 24 41	S-19/S-20/S-21(46.0-57.5) PL=NP MC=19.7 USCS=SP-SM P#200=11.7		46	46.0' - EL -31.1 Poorly graded SAND with silt (SP-SM); sand is fine to coarse; rounded; nonplastic; gray, greenish; wet; NP; medium dense to very dense; homogeneous.	GLACIAL.
47.5	A-N					47		
						48		
						49		
						50		
51.0	S-20	100%	9 19 26			51		
52.5	A-N					52		
						53		
						54		
						55		
56.0	S-21	67%	7 11 16			56		
57.5	A-N					57		
						58		
						59		
						60		

Boring backfilled with cement-bentonite grout.

TEMPLATE=10L_BORING PROJECT=MULPOINT SMIG PRI LIBRARY GPJ LIBRARY=BAKER 10-1 GLB DATE PRINTED= 3/18/24 S.O.#=197889 CHECKED BY: JML

LOG OF BORING NO.: MB-1A

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 4 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 1/30/2024 END: 2/2/2024
NORTH: 5239136.236 | EAST: 726697.1227 | COORD. DATUM: UTM21-North | LOGGER: Dylan Brown
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 11 ℄ CAVED @ ___ TIME: 12:10 DATE: 2/2/2024 DESC.: 0 Hr. PLUNGE: -90
▼ H_2O @ 11.4 ℄ CAVED @ ___ TIME: 13:30 DATE: 2/3/2024 DESC.: 25.5 Hr. ELEV. DATUM: CGVD2013
▼ H_2O @ ___ ℄ CAVED @ ___ TIME: ___ DATE: ___ DESC.: ___ GROUND ELEV.: 14.9

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
61.0	S-22	100%	16 21 29	S-22(61.0-62.5) PPen=>4.5 MC=13.6; S-22 to S-25(61.0-77.5) PL=13 LL=22 USCS=CL P#200=82.2		60	61.0' - EL -46.1 LEAN CLAY with sand (CL); sand is fine to coarse; rounded; low plasticity; gray, moist to wet; hard; homogeneous;	GLACIOMARINE.
62.5	A-N					61		
						62		
						63		
						64		
						65		
66.0	S-23	87%	19 24 30	S-23(66.0-67.5) PPen=>4.5 MC=16.8		66		
67.5	A-N					67		
						68		
						69		
						70		
71.0	S-24	100%	10 18 24	S-24(71.0-72.5) PPen=>4.5 MC=14		71		
72.5	A-N					72		
						73		
						74		
						75		
76.0	S-25	100%	22 45 38	S-25(76.0-77.5) PPen=4.5 MC=16.5		76	Increased sand content at 76.6'	
77.5	A-N					77		
						78		
						79		
						80		

Boring backfilled with cement-bentonite grout.

TEMPLATE=10L_BORING PROJECT=MULPOINT SMIG PRI LIBRARY GPJ LIBRARY=BAKER 10-1 GLB DATE PRINTED= 3/18/24 S.O.#=197889 CHECKED BY: JML

LOG OF BORING NO.: MB-1A

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 5 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 1/30/2024 END: 2/2/2024
NORTH: 5239136.236 | EAST: 726697.1227 | COORD. DATUM: UTM21-North | LOGGER: Dylan Brown
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 11 ℄ CAVED @ ___ TIME: 12:10 DATE: 2/2/2024 DESC.: 0 Hr. PLUNGE: -90
▼ H_2O @ 11.4 ℄ CAVED @ ___ TIME: 13:30 DATE: 2/3/2024 DESC.: 25.5 Hr. ELEV. DATUM: CGVD2013
▼ H_2O @ ___ ℄ CAVED @ ___ TIME: ___ DATE: ___ DESC.: ___ GROUND ELEV.: 14.9

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
81.0	S-26	75%	50/0.4			80	81.0' - EL -66.1 Poorly-graded GRAVEL with sand (gp); sand is fine to coarse; gravel is coarse; subangular; nonplastic; gray, greenish; wet; NP; very dense; homogeneous. Evidence of cobbles and boulders.	GLACIAL.
81.4	A-N					81		
						82		
						83		
						84		
						85		
86.0	S-27	100%	30 50/0.1			86		
86.6	A-N					87		
						88		
						89		
						90		
91.0	S-28	53%	13 35 50			91		
92.5	A-N					92		
						93		
						94		
						95		
96.0	S-29	100%	50/0.4			96		
96.4	A-N					97		
						98		
						99		
						100	100.1' - EL -85.2	

Boring backfilled with cement-bentonite grout.

TEMPLATE=10L_BORING PROJECT=MULPOINT SMIG PRI LIBRARY GPJ LIBRARY=BAKER 10-1 GLB DATE PRINTED= 3/18/24 S.O.#=197889 CHECKED BY: JML

APPR.	DATE	DESCRIPTION	STM

NAVFAC

SEAL

A/E INFO

APPROVED: Wm B Forbes — Digitally signed by signature Date: 2026.01.28 16:52:18-05'00'
FOR COMMANDER NAVFAC

ACTIVITY: D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026

DES LAP | DRW LAP | CHK CFC
PM/DM MJD/ATP
BRANCH MANAGER
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NORFOLK VA
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
SUBSURFACE EXPLORATION RECORDS

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157073
SHEET 11 OF 170
B-601

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\CONUS\~MultipleSites\2026_1783264_P1360_MJMF\B_Design\DWGS\Sheet\04_GEO\P1360-1783264-B-601.dwg LAYOUT NAME: B-601 PLOTTED: Wednesday, January 21, 2026 - 8:07am USER: tara.patton

LOG OF BORING NO.: MB-1A

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 6 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 1/30/2024 END: 2/2/2024
NORTH: 5239136.236 | EAST: 726697.1227 | COORD. DATUM: UTM21-North | LOGGER: Dylan Brown
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 11 ∠ CAVED @ | TIME: 12:10 | DATE: 2/2/2024 | DESC.: 0 Hr. | PLUNGE: -90
▼ H_2O @ 11.4 ∠ CAVED @ | TIME: 13:30 | DATE: 2/3/2024 | DESC.: 25.5 Hr. | ELEV. DATUM: CGVD2013
▼ H_2O @ ∠ CAVED @ | TIME: | DATE: | DESC.: | GROUND ELEV.: 14.9

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
100.0	S-30	100%	50/0.1			100	End of Boring at 100.1'	
100.1						101		
						102		
						103		
						104		
						105		
						106		
						107		
						108		
						109		
						110		
						111		
						112		
						113		
						114		
						115		
						116		
						117		
						118		
						119		
						120		

Boring backfilled with cement-bentonite grout.

CHECKED BY: JML
TEMPLATE=15L_BORING_PROJECT=MULPOINT SMIG PRI LIBRARY.GPJ LIBRARY=BAKER_15-1.GLB DATE PRINTED= 3/18/24 S.O.#=197889

LOG OF BORING NO.: MB-1B

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 1 OF 1
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 2/5/2024 END: 2/5/2024
NORTH: 5239138.503 | EAST: 726697.7275 | COORD. DATUM: UTM21-North | LOGGER: John Laux
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: 4 in OD solid auger

▽ H_2O @ Dry ∠ CAVED @ | TIME: 12:50 | DATE: 2/5/2024 | DESC.: 0 Hr. | PLUNGE: -90
▼ H_2O @ ∠ CAVED @ | TIME: | DATE: | DESC.: | ELEV. DATUM: CGVD2013
▼ H_2O @ ∠ CAVED @ | TIME: | DATE: | DESC.: | GROUND ELEV.: 14.9

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
0.0				Bag 1(0.0-10.0) PL=22 LL=23 MC=16.5 USCS=SM P#200=16.5		0	Silty SAND with gravel (SM); sand is fine to coarse; gravel is fine to coarse; nonplastic; moist; NP; heterogeneous.	
	A-N					1		
						2		
						3		
						4	Environmental sample ID: 24020503 collected at 4.0'-7.0'	
						5		
						6		
						7		
						8	Environmental sample ID: 24020504 collected at 7.0'-10.0'	
						9		
10.0						10	10.0' - EL 4.9 End of Boring at 10.0'	

Offset borehole to collect enviromental samples within the sample zone that had low recovery at the original borehole location. Boring backfilled with cuttings and cement bentonite grout.

CHECKED BY: JML
TEMPLATE=15L_BORING_PROJECT=MULPOINT SMIG PRI LIBRARY.GPJ LIBRARY=BAKER_15-1.GLB DATE PRINTED= 3/18/24 S.O.#=197889

LOG OF BORING NO.: MB-2

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 1 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 2/2/2024 END: 2/5/2024
NORTH: 5239158.145 | EAST: 726688.3345 | COORD. DATUM: UTM21-North | LOGGER: John Laux
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 9.9 ∠ CAVED @ | TIME: 11:00 | DATE: 2/5/2024 | DESC.: 0 Hr. | PLUNGE: -90
▼ H_2O @ 10.3 ∠ CAVED @ | TIME: 08:55 | DATE: 2/6/2024 | DESC.: 22 Hr. | ELEV. DATUM: CGVD2013
▼ H_2O @ ∠ CAVED @ | TIME: | DATE: | DESC.: | GROUND ELEV.: 14.2

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
0.0						0	TOPSOIL. 0.3' - EL 13.9	FILL.
	S-1	60%	2 15 29			1	Poorly graded GRAVEL with silt and sand (GP-GM); sand is fine to coarse; gravel is fine to coarse, angular; nonplastic; gray, brownish; moist; NP; loose to dense; heterogeneous.	FILL.
1.5						2		
	S-2	73%	24 27 19	S-2 to S-6(1.5-9.0) PL=NP MC=9.7 USCS=GP-GM P#200=9.4		3		
3.0						4		
	S-3	47%	15 21 22			5		
4.5				S-4(4.5-6.0) PID=26 ppm		6	Chunks of wood in S-4	
	S-4	20%	18 23 5			7		
6.0						8		
	S-5	0%	5 5 2			9		
7.5						10		
	S-6	27%	4 2 4			11		
9.0						12	12.0' - EL 2.2	
	S-7	0%	4 4 4			12	Poorly graded GRAVEL with silt and sand (GP-GM); sand is fine to coarse; gravel is fine to coarse; subangular; nonplastic; gray, greenish; moist to wet; NP; medium dense to very dense; heterogeneous.	GLACIAL.
10.5						13		
	S-8	47%	3 20 18			14		
12.0						15		
12.1	S-9	0%	50/0.1			16		
	A-N					17		
13.5						18		
	S-10	33%	25 37 50/0.2			19		
14.7	A-N					20		
15.0				Bag 1(15.0-71.0) PL=NP MC=17.8 USCS=SM P#200=17.5			Cored through large gravel/boulders 15-18'	
	S-11	47%	13 16 14					
16.5				S-11 to S-14(15.0-21.0) PL=NP MC=7.6 USCS=GP-GM P#200=7.3				
	S-12	33%	28 17 24					
18.0								
	S-13	13%	20 21 20					
19.5								

Boring backfilled with cement-bentonite grout.

CHECKED BY: JML
TEMPLATE=15L_BORING_PROJECT=MULPOINT SMIG PRI LIBRARY.GPJ LIBRARY=BAKER_15-1.GLB DATE PRINTED= 3/18/24 S.O.#=197889

NOTES

1. 1 METER = 3.28 FEET

LOG OF BORING NO.: MB-2

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 2 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 2/2/2024 END: 2/5/2024
NORTH: 5239158.145 | EAST: 726688.3345 | COORD. DATUM: UTM21-North | LOGGER: John Laux
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 9.9 ∠ CAVED @ | TIME: 11:00 | DATE: 2/5/2024 | DESC.: 0 Hr. | PLUNGE: -90
▼ H_2O @ 10.3 ∠ CAVED @ | TIME: 08:55 | DATE: 2/6/2024 | DESC.: 22 Hr. | ELEV. DATUM: CGVD2013
▼ H_2O @ ∠ CAVED @ | TIME: | DATE: | DESC.: | GROUND ELEV.: 14.2

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
21.0	S-14	20%	17 14 16			20		
	A-N					21		
						22		
						23		
						24		
						25		
26.0	S-15	20%	3 14 28			26		
27.5	A-N					27		
						28		
						29		
31.0	S-16	100%	31 50/0.3			30		
31.8	A-N					31	Increased sand content beginning at 31.0	
						32		
						33		
						34		
						35		
36.0	S-17	20%	1 1 13			36	36.0' - EL -21.8 Poorly graded SAND with gravel (sp); sand is fine to coarse; gravel is fine, subangular; nonplastic; gray, greenish; wet; NP; loose to very dense; homogeneous.	GLACIAL.
37.5	A-N					37		
						38		
						39		
						40		

Boring backfilled with cement-bentonite grout.

CHECKED BY: JML
TEMPLATE=15L_BORING_PROJECT=MULPOINT SMIG PRI LIBRARY.GPJ LIBRARY=BAKER_15-1.GLB DATE PRINTED= 3/18/24 S.O.#=197889

LOG OF BORING NO.: MB-2

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 3 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 2/2/2024 END: 2/5/2024
NORTH: 5239158.145 | EAST: 726688.3345 | COORD. DATUM: UTM21-North | LOGGER: John Laux
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 9.9 ∠ CAVED @ | TIME: 11:00 | DATE: 2/5/2024 | DESC.: 0 Hr. | PLUNGE: -90
▼ H_2O @ 10.3 ∠ CAVED @ | TIME: 08:55 | DATE: 2/6/2024 | DESC.: 22 Hr. | ELEV. DATUM: CGVD2013
▼ H_2O @ ∠ CAVED @ | TIME: | DATE: | DESC.: | GROUND ELEV.: 14.2

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
41.0						40		
	S-18	93%	10 22 37			41		
42.5						42		
	A-N					43		
						44		
						45		
46.0	S-19	67%	23 25 23			46		
47.5	A-N					47		
						48		
						49		
51.0	S-20	0%	3 3 5			50		
52.5	A-N					51		
						52		
						53		
						54		
56.0	S-21	100%	25 32 41			55	Sand became more fine from 56-66	
57.5	A-N					56		
						57		
						58		
						59		
						60		

Boring backfilled with cement-bentonite grout.

CHECKED BY: JML
TEMPLATE=15L_BORING_PROJECT=MULPOINT SMIG PRI LIBRARY.GPJ LIBRARY=BAKER_15-1.GLB DATE PRINTED= 3/18/24 S.O.#=197889

LOG OF BORING NO.: MB-2

Michael Baker INTERNATIONAL

PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services | SHEET: 4 OF 6
LATITUDE: | LONGITUDE: | GEO. DATUM: NAD83 | START: 2/2/2024 END: 2/5/2024
NORTH: 5239158.145 | EAST: 726688.3345 | COORD. DATUM: UTM21-North | LOGGER: John Laux
STATION: | OFFSET: | BASELINE: | DRILLER: Brendan Clark
EQUIPMENT: CME 55 Marooka | DRILL CO.: Logan Geotech
METHOD DETAILS: HQ series core, w/ water. 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▽ H_2O @ 9.9 ∠ CAVED @ | TIME: 11:00 | DATE: 2/5/2024 | DESC.: 0 Hr. | PLUNGE: -90
▼ H_2O @ 10.3 ∠ CAVED @ | TIME: 08:55 | DATE: 2/6/2024 | DESC.: 22 Hr. | ELEV. DATUM: CGVD2013
▼ H_2O @ ∠ CAVED @ | TIME: | DATE: | DESC.: | GROUND ELEV.: 14.2

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	SPT BLOWS/0.5 FT or (RQD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
61.0						60		
	S-22	100%	19 22 29			61		
62.5						62		
	A-N					63		
						64		
						65		
66.0	S-23	100%	7 10 3			66		
67.5	A-N					67		
						68		
						69		
71.0	S-24	100%	2 3 3			70		
72.5	A-N					71		
						72	72.2' - EL -58.0 SILTY CLAY with sand (CL-ML); sand is fine to coarse, subrounded; low plasticity; gray; wet; very stiff to hard; homogeneous.	GLACIOMARINE. S-26 and S-27 extra sample in bags
						73		
						74		
76.0	S-25	53%	40 27 30	S-25(S-26(76.0-82.5) PL=15 LL=21 MC=16.4 USCS=CL-ML P#200=78.9		75		
77.5	A-N					76		
						77		
						78		
						79		
						80		

Boring backfilled with cement-bentonite grout.

CHECKED BY: JML
TEMPLATE=15L_BORING_PROJECT=MULPOINT SMIG PRI LIBRARY.GPJ LIBRARY=BAKER_15-1.GLB DATE PRINTED= 3/18/24 S.O.#=197889

STM	DESCRIPTION	DATE	APPR.

NAVFAC

SEAL

A/E INFO

APPROVED: Wm B Forbes — Digitally signed by signature Date: 2026.01.28 16:52:18-05'00'
FOR COMMANDER NAVFAC

ACTIVITY: D. EVANS VIA EMAIL
SATISFACTORY TO DATE: 01/15/2026
DES: LAP | DRW: LAP | CHK: CFC
PM/CM: MJD/ATP
BRANCH MANAGER:
DES PROG DIR: WBF
FIRE PROTECTION ENGINEER: DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION
NORFOLK VA
VARIOUS LOCATIONS
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
SUBSURFACE EXPLORATION RECORDS

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157074
SHEET 12 OF 170
B-602

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\OCONUS\~Multiple Sites\2026_1783264_P1360_MJMF\B_Design\DWGS\Sheet\04_GEO\P1360-1783264-B-601.dwg LAYOUT NAME: B-602 PLOTTED: Wednesday, January 21, 2026 - 8:07am USER: tara.patton